

12. A student was provided with five ionic solids containing familiar cations and anions. Each solid contains a different **cation**. The solids were labelled **A**, **B**, **C**, **D** and **E**.

The student attempted to dissolve all five solids in water. She then attempted to dissolve those that did not dissolve in water in a stoichiometric amount of acid.

The results are given below.

Solid	Addition of water	Addition of dilute sulfuric acid to solid	Addition of dilute nitric acid to solid
A	dissolves giving colourless solution 1		
B	dissolves giving colourless solution 2		
C	does not dissolve	pale blue solution 3 forms upon warming with the acid	
D	does not dissolve	effervescence and solution 4 forms	
E	does not dissolve	some effervescence but the solid does not dissolve	effervescence and solution 5 forms

Pairs of the solutions formed were mixed and the observations recorded.

	Solution 5	Solution 4	Solution 3	Solution 2
Solution 1 (formed by dissolving solid A)	no visible change	white precipitate	white precipitate	bright yellow precipitate
Solution 2 (formed by dissolving solid B)	no visible change	no visible change	brown solution with a white solid	
Solution 3 (formed by reacting solid C with acid)	thick white precipitate	no visible change		
Solution 4 (formed by reacting solid D with acid)	thick white precipitate			

Flame tests

Flame tests carried out on solutions **1**, **2**, **3**, **4** and **5** gave no colour with one solution, an apple-green flame with another and a lilac flame with a third. The student noted unfamiliar flame test colours for the other two solutions.



